

Ref Here's a more raw, less "polished AI" sounding version of your Lab 11 reflective journal, Mary Ann — with more frustration, learnings, and reflection that feels like you actually wrote it after wrestling with the lab:

Reflective Journal – Lab 11: Advanced Tracking and System Design

Overview

Lab 11 built on Lab 10 but pushed things further into system design and evaluation. Instead of just noticing when trackers failed, I had to measure their performance with metrics and think about how they'd actually work in real environments. It wasn't just about running code — it was about making sense of messy outputs and realizing that "better" isn't always clear-cut.

Key Learnings

- **Performance Metrics:** Using ID accuracy, track length, and fragmentation gave me a way to judge trackers beyond just eyeballing them. Honestly, it was tedious at first, but it forced me to stop relying on gut feelings.
- **Failure Diagnosis:** The same failure modes from Lab 10 showed up again, but in more crowded videos they were even worse. ID switches stacked up quickly, and phantom tracks made me roll my eyes because they looked so obviously wrong.
- **System Constraints:** Seeing how GPU cost, bandwidth, and latency affect tracker choice made me realize why engineers can't just pick the "best" algorithm. Sometimes you have to settle for what runs fast enough.
- **Real-World Application:** I started thinking about industries like traffic monitoring or retail analytics, where accuracy matters but resources are limited. It's not about perfection — it's about what's good enough to be useful.

Application to Practice

This lab made me think about how trackers would actually be deployed. In traffic systems, motion-based trackers might be fine because cars move predictably. In crowded retail spaces, appearance cues might be necessary to avoid ID switches when people overlap. In sports, fragmentation could ruin analysis if a player's track keeps breaking. It's not about chasing the most advanced model — it's about matching the tool to the environment and constraints.

Reflection

I'll be honest: this lab was frustrating trying to count all those people. Metrics felt like busywork at first, but then I realized they were the only way to prove whether a tracker was working. Just watching videos wasn't enough — I had to quantify the failures.

The crowded videos were the worst. Trackers broke down and I was having difficulty trying to count all the people when IDs switched or phantom tracks popped up. But I did notice patterns: ByteTrack struggled when motion slowed, BoT-SORT handled overlaps better but ate more

resources. It was messy, but it made me think like an engineer instead of just a student running code.

The biggest takeaway is that tracking isn't a solved problem. It's a balancing act between accuracy, speed, and resources, and there's no perfect answer. Lab 11 forced me to accept that sometimes "good enough" is the real goal, and that designing systems means living with trade-offs. It wasn't a clean lab, but it was one of the most realistic — and that's what made it valuable. I still have no idea exactly how many people were in there and gave it my best shot in spite of the bounding boxes were getting in the way of seeing and counting people.